

Technical paper

A vision-enabled fatigue-sensitive human digital twin towards human-centric human-robot collaboration

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ABSTRACT

Within a Human-centric Human-Robot Collaboration (HHRC) system, monitoring, assessing, and optimizing for an operator's well-being is essential to creating an efficient and comfortable working environment. Currently, monitoring systems are used for independent assessment of human factors. However, the rise of the Human Digital Twin (HDT) has provided the framework for synchronizing multiple operator well-being assessments to create a comprehensive understanding of the operator's performance and health. Within manufacturing, an operator's dynamic well-being can be attributed to their physical and cognitive fatigue across the assembly process. As such, we apply non-invasive video understanding techniques to extract relevant assembly process information for automatic physical fatigue assessment. Our novelty involves a video-based fatigue estimation method, in which the boundary-aware dual-stream MS-TCN combined with an LSTM is proposed to detect the operation type, operation repetitions, and the target arm performing each task in an assembly process video. The detected results are then input into our physical fatigue profile to automatically assess the operator's localized physical fatigue impact. The assembly process of a real-world bookshelf is recorded and tested against, with our algorithm results showing superiority in operation segmentation and target arm detection as opposed to other recent action segmentation models. In addition, we integrate a cognitive fatigue assessment tool that captures operator physiological signals in real-time for body response detection caused by stress. This provides a more robust HDT of the operator for an HHRC system.

1. Introduction

Human-Robot Collaboration (HRC) is a potential solution to the need for a versatile, ultra-flexible, and self-adaptive manufacturing system as the shift toward mass personalization transitions the manufacturing shop floor toward an unstructured, dynamic environment [1,2]. An operator's cognition, flexibility, and adaptability are paired with a robot's precision, strength, and repeatability to create a collaborative partnership that increases productivity beyond either agent alone [2]. Traditional HRC is system-centric, restricting operators to predefined roles matching their robot partners to optimize a profit-driven objective. Alternatively, the Industry 5.0 initiative [3] and human-centric manufacturing research [4,5] emphasize the shift from system-centric manufacturing to more sustainable and resilient, human-centric manufacturing. Thus, HRC is preferred to evolve into Human-centric Human-Robot Collaboration (HHRC) [5], which prioritizes operator well-being alongside productivity within the system objectives.

To ensure high-quality collaboration within an HHRC system, it is

essential to understand an operator's dynamic state (physical, cognitive, and psychological health) continuously and accurately across the manufacturing process, leading to significant research efforts into operator health assessment in manufacturing [6,7]. Particularly, monitoring an operator's dynamic physical and cognitive fatigue is critical to maintaining comfortable working conditions in manufacturing environments. Current high-accuracy real-time monitoring of physical and cognitive fatigue often involves restrictive equipment, such as Surface-Electromyography (s-EMG) [8,9] or Electroencephalography (EEG) technologies, that can inhibit an operator's full range-of-motion and cause discomfort. Furthermore, the high cost and privacy concerns associated with such intrusive sensing methods pose additional challenges in industry adoption, leading emerging ideas to explore vision-based and wrist wearable sensors as an alternative [10,11]. Although current physical and cognitive fatigue assessment methods have shown promising results in manufacturing, most research often consider isolated assessments of one or few related factors, leaving an operator's holistic and robust modeling largely unexamined.

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Creating a Human Digital Twin (HDT) is a promising solution to providing a holistic and encompassing model of an operator's dynamic state [12,13]. An HDT is a digital replica of the physical operator, where real-time monitoring and sensing lead to high-fidelity synchronization between the physical environment and a virtual construct, creating an advantageous position for optimizing system performance and operator well-being via intelligent and flexible deployment of HRC resources (such as task allocation and robot control schemes). Nevertheless, an HDT that encompasses an operator's dynamic physical and cognitive states for HHRC application is still lacking.

Given this, we propose an HDT framework for modeling and monitoring both physical and cognitive fatigue of operators, towards Human-centric Human-Robot Collaboration. On physical fatigue, the HDT uses our unique video-based human action segmentation techniques to monitor an operator's localized muscular fatigue in both arms during prolonged operations simultaneously. On cognitive fatigue, a light wearable sensor (smartwatch) is incorporated into the system to recognize an operator is cognitively fatigued by continuously tracking their physiological signals (heart rate, breathing rate, and skin temperature).

Our knowledge contributions include the following:

1. We propose a Human Digital Twin framework that models both physical and cognitive fatigue of operators throughout an assembly process. This framework is seamlessly integrated into a Human-centric Human-Robot Collaboration system, enabling holistic understanding of operator state and performance.
2. We design a vision-based boundary aware operation segmentation algorithm capable of classifying repetitive operations performed by an operator using either arm for each task across an assembly process video. We also couple this vision-based monitoring system with our previously established localized muscle fatigue profile for personalized, non-contact physical fatigue assessment.
3. We incorporated a lightweight smartwatch to monitor cognitive fatigue via real-time changes in physiological signals during dynamic manufacturing operations.
4. We extensively evaluate the feasibility and effectiveness of our proposed fatigue (both physical and cognitive) monitoring and assessment approach for assembling a real-world product involving 26 tasks and six unique operation types.

The remainder of this paper is as follows. Section 2 reviews related literature. Section 3 overviews our proposed HDT framework, while Section 4 details the physical and cognitive fatigue monitoring solutions. Section 6 evaluates the performance of our solutions in a real-world assembly procedure. The paper concludes with limitations and future works in Section 6.

2. Related works

This section provides an overview of Human-centric Human-Robot Collaboration (HHRC), physical and cognitive fatigue assessment, and action segmentation techniques in video understanding. Subsequently, we outline the research gaps that motivated this study.

2.1. Human-centric human-robot collaboration

Mass personalization signifies efficiently producing highly personalized products with variable batch sizes [1]. This manufacturing paradigm necessitates highly flexible manufacturing systems adapting to diverse production requirements via leveraging enabling technologies. One such manufacturing system is HRC, which blends the cognition, flexibility, and adaptability of humans with the precision, strength, and repeatability of robots to achieve greater system performance than either agent alone [2,14].

Standard HRC is system-centric, aiming to maximize productivity by

assigning operators defined roles that align manual labor efficiency with automation [15]. However, future manufacturing has begun to embrace a working philosophy shift towards human-centric manufacturing following the recent Industry 5.0 initiative [3] and human-centric manufacturing research [4,5] - "manufacturing practices that place human wellbeing at the center of manufacturing systems and processes, providing a safe, comfortable, and motivating environment for working, learning and growth, thereby creating a virtuous circle in which individual wellbeing and sustained manufacturing growth are mutually reinforcing" [5].

Incorporating similar principles, HRC was preferred to evolve into HHRC [5]. In place of system-centric objectives, HHRC optimizes human-centric objectives to prioritize operator well-being alongside system efficiency. Yet, before optimization can occur, the system must be able to continuously monitor an operator's dynamic state (physical, cognitive, and psychological health) within the flexible manufacturing environment. Creating an HDT that utilizes multimodal sensing to perceive an operator's dynamic state is critical for high-fidelity synchronization between the physical environment and the system's virtual construct. Specifically, looking at an operator's dynamic physical and cognitive fatigue is paramount to creating a safe, comfortable, and productive working environment.

2.2. Dynamic physical and cognitive fatigue modeling

Modeling an operator's dynamic state (physical, cognitive, and psychological health) is vital to an effective HHRC system, where an operator's needs often fluctuate during continuous manufacturing operations. Notably, physical [16] and cognitive fatigue are leading factors of workplace injuries and reduced system performance.

Dynamic physical fatigue, best represented by localized muscle fatigue, is the temporary and localized sensation of a target muscle's reducing capacity to generate force due to strenuous and repetitive physical activity that overburdens the target muscle [16]. Researchers primarily use s-EMG (surface electromyography) sensors to measure the myoelectric signal changes resulting from fatigue accumulation in the target muscle, where a spectral shift to lower frequencies is observed as muscle fiber conduction velocity decreases due to muscle exhaustion [8]. Spectral fatigue indicators from the target muscle's myoelectric signal have been extensively employed to describe fatigue accumulation from static, isometric, and dynamic, single-degree-of-freedom contractions. However, the primary limitations of s-EMG sensing include its restrictive and uncomfortable nature for the operator, especially over prolonged use in manufacturing environments. Additionally, the accuracy of the fatigue assessment is compromised in unstructured dynamic movements as the myoelectric signal behaves abnormally when muscle fibers stretch and contract unexpectedly, thereby limiting direct s-EMG measurement in manufacturing environments.

Dynamic cognitive fatigue, also referred to as mental fatigue or stress, is a state of mental exhaustion that can significantly impair an operator's cognitive performance. Tasks that demand high precision, strict adherence to time constraints, familiarity, or those performed in noisy or distracting environments are known to induce cognitive fatigue. In clinical studies, researchers have utilized EEG to measure electrical signal changes in the brain by attaching sensors directly to an operator's scalp, facilitating detailed analysis of brain activity related to cognitive states [17,18]. Another method, Electrocardiography (ECG), is used to measure cardiac activity, offering insights into heart rate variability as an indicator of stress and fatigue [19,20]. Additionally, eye tracking technology is used to analyze eye movement patterns and pupil dilation, which are responsive to cognitive load and fatigue [21]. Despite their relative precision, these methods can be restrictive and uncomfortable, particularly in dynamic manufacturing environments. In response to these limitations, researchers have started exploring lightweight, wrist wearable sensors, such as such as wristbands or smartwatches, to monitor physiological signals indicative of cognitive fatigue [11,22]. These signals include heart rate variability, breathing rate, skin

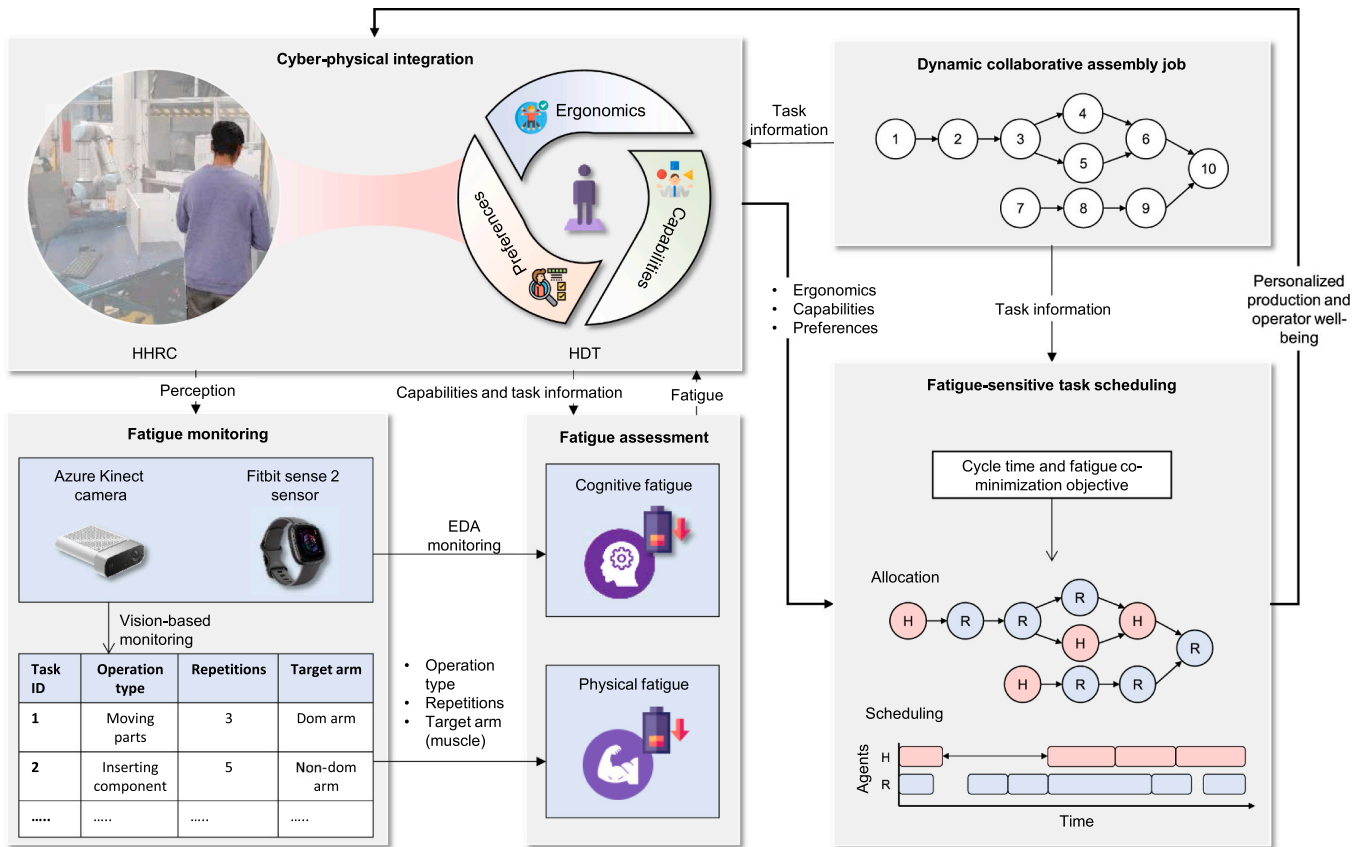


Fig. 1. Overview of our fatigue-sensitive HHRC system.

temperature, and Electrodermal Activity (EDA). Wrist wearable sensors provide non-restrictive and effective means to detect cognitive fatigue in various operational settings.

Our recent work [23] utilized s-EMG to determine how key parameters, such as the operator's target muscle strength, operation type, repetition count, and task load, influence an operator's target muscle fatigue rate. This approach creates a personalized fatigue profile for assessing localized muscle fatigue in dynamic, multiple-degree-of-freedom manufacturing operations by calculating fatigue accumulation using s-EMG in static muscle contractions between operation sets. By creating a precise fatigue profile in a controlled setting, we developed an assessment protocol that infers localized muscle fatigue without requiring s-EMG sensors, instead relying on key observable parameters. Building on this, this paper demonstrates the potential of vision-based systems to capture all key parameters for the personalized fatigue profile via non-contact sensing. Moving forward, we refer to localized muscular fatigue as physical fatigue for brevity.

2.3. Action segmentation in video understanding

Action segmentation techniques in video analysis are designed to identify specific actions within a given video. This is crucial for extracting information on assembly processes in manufacturing, enabling the assessment of physical fatigue by identifying the target muscle, the operation type, and the operation repetitions. Early action segmentation approaches use object state changes [10,24] or sliding windows [25,26] to classify action boundaries but cannot infer context over long-time videos with multiple consecutive actions [27]. An alternative method to address this issue is a temporal model that predicts the action of each frame sequentially [28,29]. However, the model requires more computational resources as it cannot predict frames simultaneously. [30] introduced a Temporal Convolutional Network

(TCN) into action segmentation after the model's success in speech synthesis. With TCN, researchers could recognize long-term dependencies at higher efficiencies than standard temporal models. State-of-the-art action segmentation models are built on TCN, such as the Multi-Stage TCN (MS-TCN) [27], which refine action predictions progressively using multiple TCN modules. Researchers have also tried to improve the MS-TCN in subsequent studies [31,32]. However, current state-of-the-art action segmentation models cannot recognize consecutive repetitive actions as individual actions. Rather, they classify the boundary of the repeated actions as one action type, as each frame has an isolated action prediction aggregated between continuous frames of the same action.

2.4. Research gaps and motivations

Based on our brief literature review, we identify the following research gaps that motivate this study:

1. Existing methods have proven effective for assessing physical or cognitive fatigue independently in manufacturing, rather than evaluating both simultaneously. However, a robust and holistic model of an operator, derived from non-restrictive monitoring of both physical and cognitive fatigue, remains largely unexamined.
2. In physical fatigue assessment, high-accuracy and objective evaluation often demand restrictive s-EMG technologies. Our recent work [23] on developing an offline fatigue profile, based on s-EMG technologies for online use, enables automatic personalized physical fatigue inference with the necessary assembly process information provided by the system. As it stands, current action segmentation video understanding techniques are unable to reliably segment operation repetitions or identify an operator's target arm performing each task.

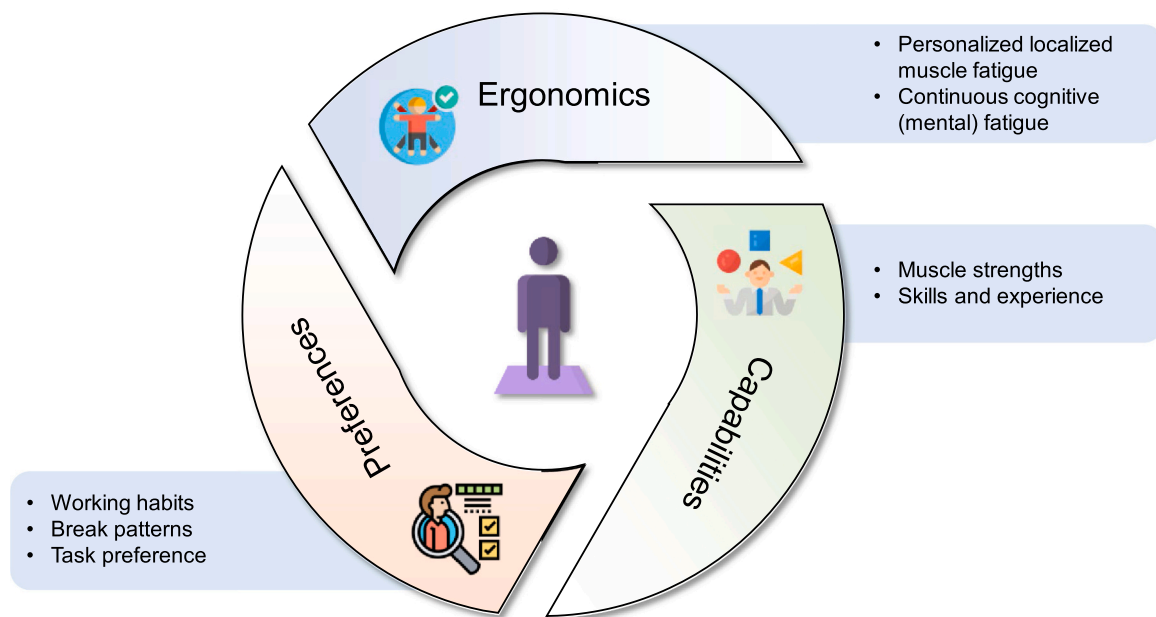


Fig. 2. Outline of our Human Digital Twin (HDT).

Motivated by the above, we model an operator's dynamic physical and cognitive fatigue across an assembly process as an HDT. This HDT framework will allow the system to proactively realign its resources to best manage the operator's changing states.

For precise monitoring of assembly processes and accurate visual estimation of physical fatigue, it is crucial to differentiate between repetitive operations (assembly actions) and identify the target muscle involved in these actions. Our previous study [33] tried addressing this challenge by introducing the dual-stream MS-TCN framework, which identified operation boundaries in isolated tasks and utilized a simplified fatigue profile to estimate the physical fatigue impact of these tasks. However, it could not identify the operator's target muscle (left or right arm) in action. In this study, we advance our initial boundary-aware MS-TCN architecture by integrating an LSTM network to visually predict the operator's target muscle. The enhanced boundary-aware MS-TCN architecture consists of two streams: the first stream predicts the operation type for each task, while the second stream delineates the operation segments. The LSTM network processes each operation segment, determining which arm the operator is using. Given the upper-limb orientation of typical manufacturing processes, this enables us to accurately identify the localized fatigue impact of each task on the operator by leveraging our previously developed physical fatigue profile [23]. This improved architecture not only refines the boundary detection and operation type prediction but also introduces a novel dimension of muscle-specific fatigue analysis, thereby offering a more comprehensive assessment of physical fatigue in assembly tasks.

Furthermore, we integrate a commercial wearable sensor that monitors an operator's physiological signals in real-time to detect physiological responses indicative of cognitive fatigue (or stress). Finally, we assess the feasibility and effectiveness of our approach during a bookshelf assembly task comprising 26 operations.

3. Fatigue-sensitive HHRC system

The conceptual framework for our fatigue-sensitive HHRC system is illustrated in Fig. 1, highlights the cyber-physical integration of an operator's HDT. The operator's HDT is tailored to their capabilities and preferences. It continuously monitors and evaluates their ergonomic condition, particularly regarding fatigue. The system non-invasively monitors physical fatigue through vision-based sensing and a

developed physical fatigue profile. Cognitive fatigue is tracked continuously using the Fitbit Sense 2 wearable sensor, which recognizes cognitive fatigue by monitoring an operator's heart rate, heart rate variability, skin temperature, and EDA.

A dynamic, collaborative assembly job is processed for task execution. It allocates tasks between operators (H) and robots (R), considering each operator's ergonomic capabilities and preferences. The scheduling is guided by human-centric principles to optimize productivity and the operator's fatigue states, both physical and cognitive. In this context, cognitive fatigue functions as a signal, prompting the system to recognize the operator's need for a break. Tasks for that individual can be dynamically reassigned to facilitate the required rest. As jobs enter the system, tasks are allocated to ensure a balanced workload, promoting personalized production and operator well-being.

3.1. Human digital twin

Cyber-physical integration exists between the operator in physical space and their HDT, containing a digital representation of the operator's ergonomics, capabilities, and preferences. An outline of the HDT considered in this study is shown in Fig. 2.

In this study, the ergonomics of an operator encompass both their fluctuating physical and cognitive states, predominantly represented through metrics of fatigue. These states necessitate continuous, real-time monitoring and assessment to adapt to the demands typical of collaborative assembly environments. Conversely, static ergonomics, which include factors such as posture, are integrated into the system during the initial design phase of workstations but are not actively monitored.

An operator's capabilities, including their skill set, experience, and muscle strength, serve as determinants for task execution and are essential for assessing physical fatigue. While these capabilities are treated as static metrics, they incrementally evolve as operators engage in continuous work and skill development. The HDT system can update an operator's capabilities based on accumulated task time and efficiency metrics, reflecting their growth in proficiency for particular tasks. However, the integration of new capabilities that enable operators to undertake additional tasks—previously outside their skill set—due to upskilling or other forms of training remains a manual process. This manual update directly influences the task scheduling module, which

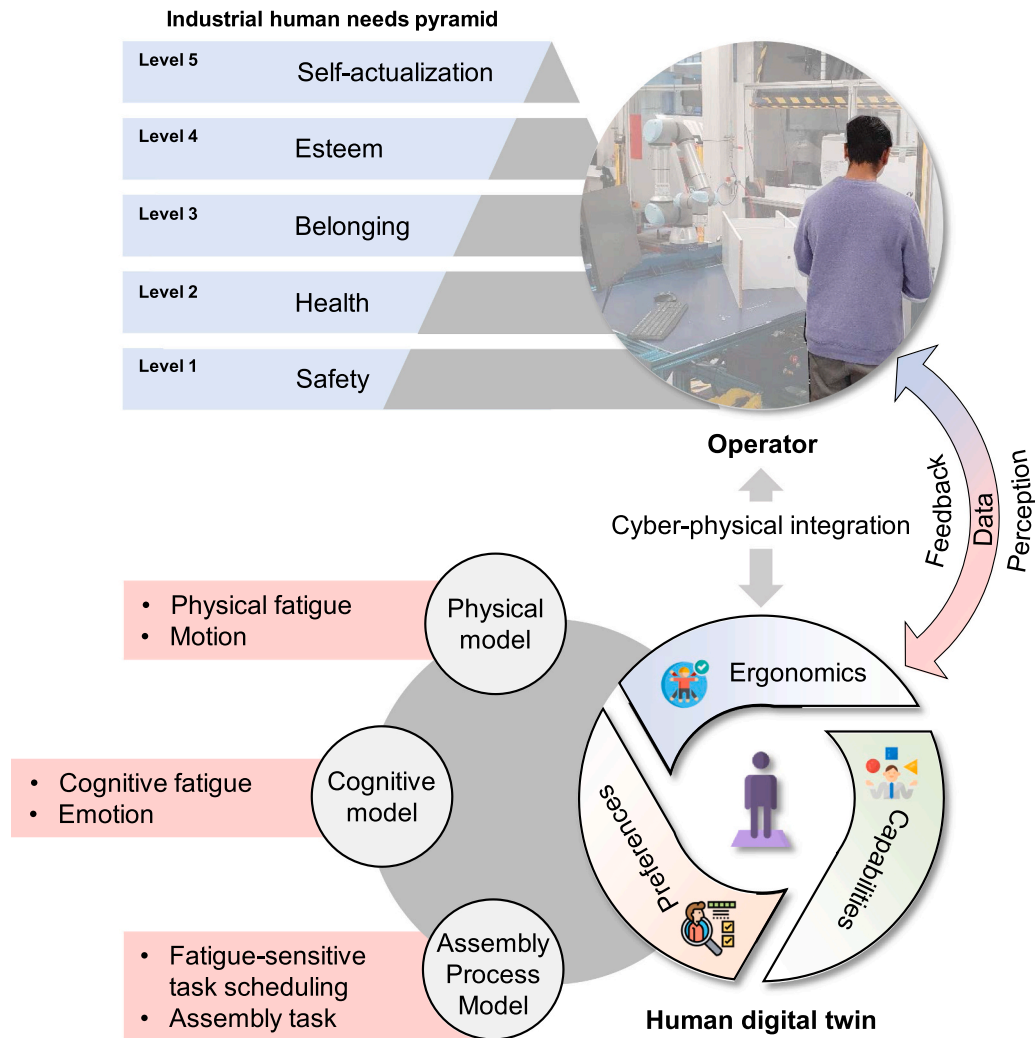


Fig. 3. Cyber-physical integration between an operator and their HDT.

relies on current capability metrics to allocate tasks appropriately.

Operator preferences encompassing working habits, break patterns, and task inclinations, are incorporated to enhance overall comfort and system efficiency. Determined by the operators themselves, these preferences ensure accurate reflection of individual inclinations and working styles. To prevent potential conflicts that could disrupt system functionality, preferences are considered passively. In scenarios where operator preferences collide, such as multiple operators favoring or avoiding specific tasks, preferences are managed as tasks that operators are disinclined to perform unless necessary. This approach helps maintain smooth operation while accommodating individual preferences.

3.2. Cyber-physical integration

Within the system, each operator in physical space is represented by a digital twin in virtual space, known as the HDT, as discussed in Section 3.1. It combines static and dynamic information of the operator throughout the assembly process. To create the HDT, we first define the operator's needs using the industrial human needs pyramid [5]. The industrial human needs pyramid contains five levels, classifying the operator's state and working habits from safety to self-actualization. In virtual space, the HDT contains three models: The physical model, the cognitive model, and the assembly process model. An overview of the cyber-physical integration is shown in Fig. 3.

The physical model captures and assesses the operator's physical state, emphasizing physical fatigue and motion. Simultaneously, the

cognitive model focuses on the operator's cognitive condition, continuously tracking their cognitive fatigue and emotional state. Complementing these, the assembly process model relates to the specific tasks the operator performs, incorporating fatigue-sensitive task scheduling to optimize the workflow based on the operator's current physical and cognitive state.

Cyber-physical integration is the bidirectional process through which data and feedback flow between the operator and the HDT. It involves dynamically collecting the operator's physical and cognitive data (perception) and providing insights from the HDT's data back to the operator (cognition).

4. Physical and cognitive fatigue monitoring

This section details our solution to developing non-invasive physical and cognitive fatigue monitoring and assessment. Our novelty in this research area primarily focuses on advancing action segmentation video understanding techniques to extract task information for physical fatigue assessment using our previous physical fatigue profile work in [23]. Additionally, we include a cognitive fatigue assessment module that detects high-stress levels (cognitive fatigue) via real-time physiological signal monitoring. An overview of the monitoring technologies and methods for fatigue assessment is illustrated in Fig. 4. The rest of this section details the monitoring and assessment of physical and cognitive fatigue.

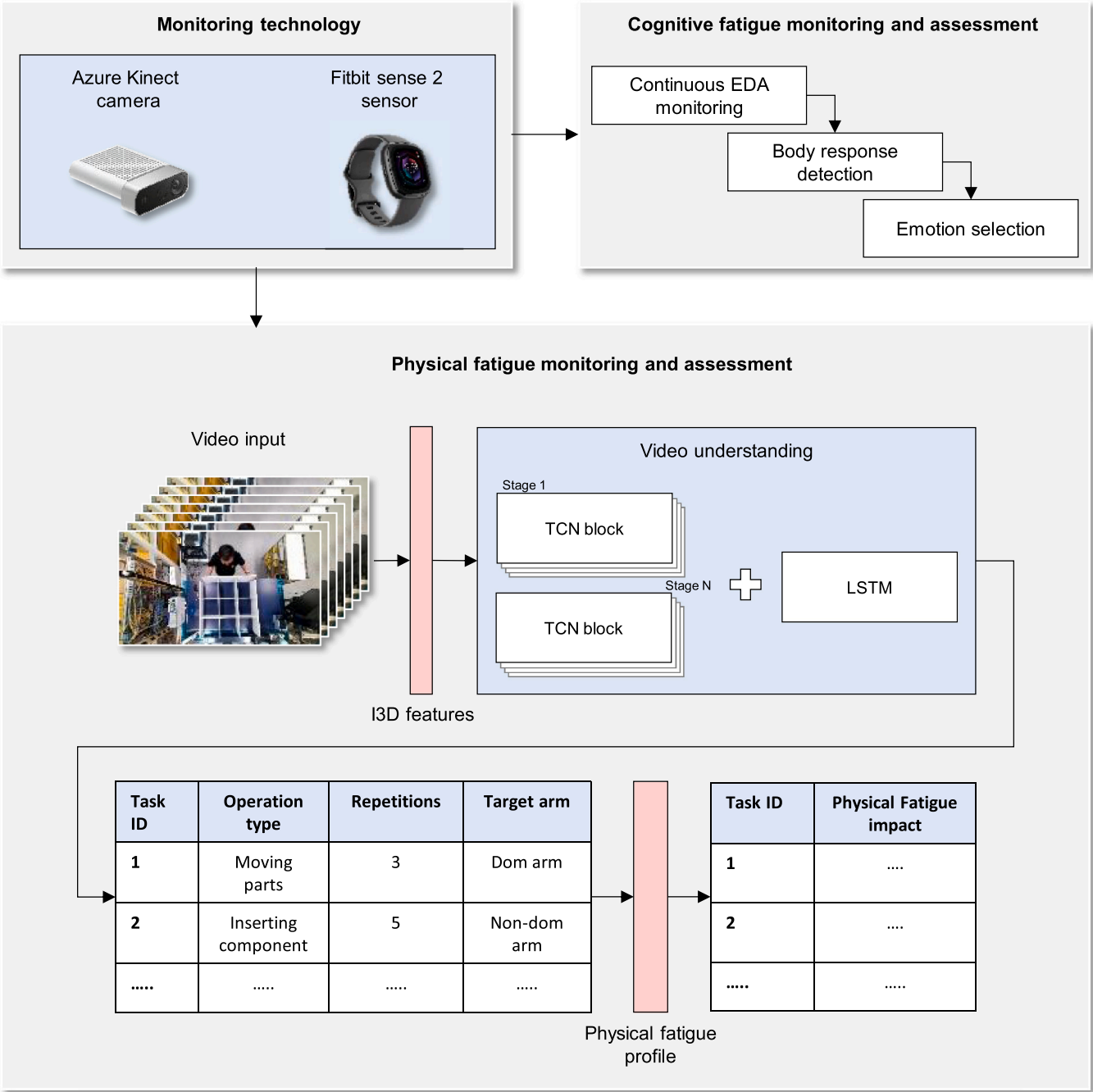


Fig. 4. Monitoring technologies and methods for physical and cognitive fatigue assessment in the HHRC system using vision-based and wearable sensors.

4.1. Physical fatigue monitoring and assessment

During the physical fatigue monitoring and assessment, a video of the assembly process is given as input, and the physical fatigue impact of each task on the operator in the assembly process is given as output. From our previous work [23], we developed a physical fatigue profile capable of quantifying the fatigue impact of assembly operations on an operator’s target arm, provided the task information (i.e., operation type, number of repetitions, and task load). From here, it becomes possible to assess physical fatigue non-invasively by visually observing the task information using action segmentation video understanding techniques.

4.1.1. Boundary-aware dual-stream MS-TCN with an LSTM

Our boundary-aware dual-stream MS-TCN with LSTM algorithm serves to extract three critical features from the assembly process:

- The operation type of each task.
- The number of operation repetitions for that task (operation segments).
- The target muscle (either left or right arm) that is performing the task.

Current MS-TCN algorithms can segment an assembly process into a series of operations that differ between operation types. As a framewise classifier, the algorithm can link between the current frame and the predicted operation, grouping frames of the same operation into an

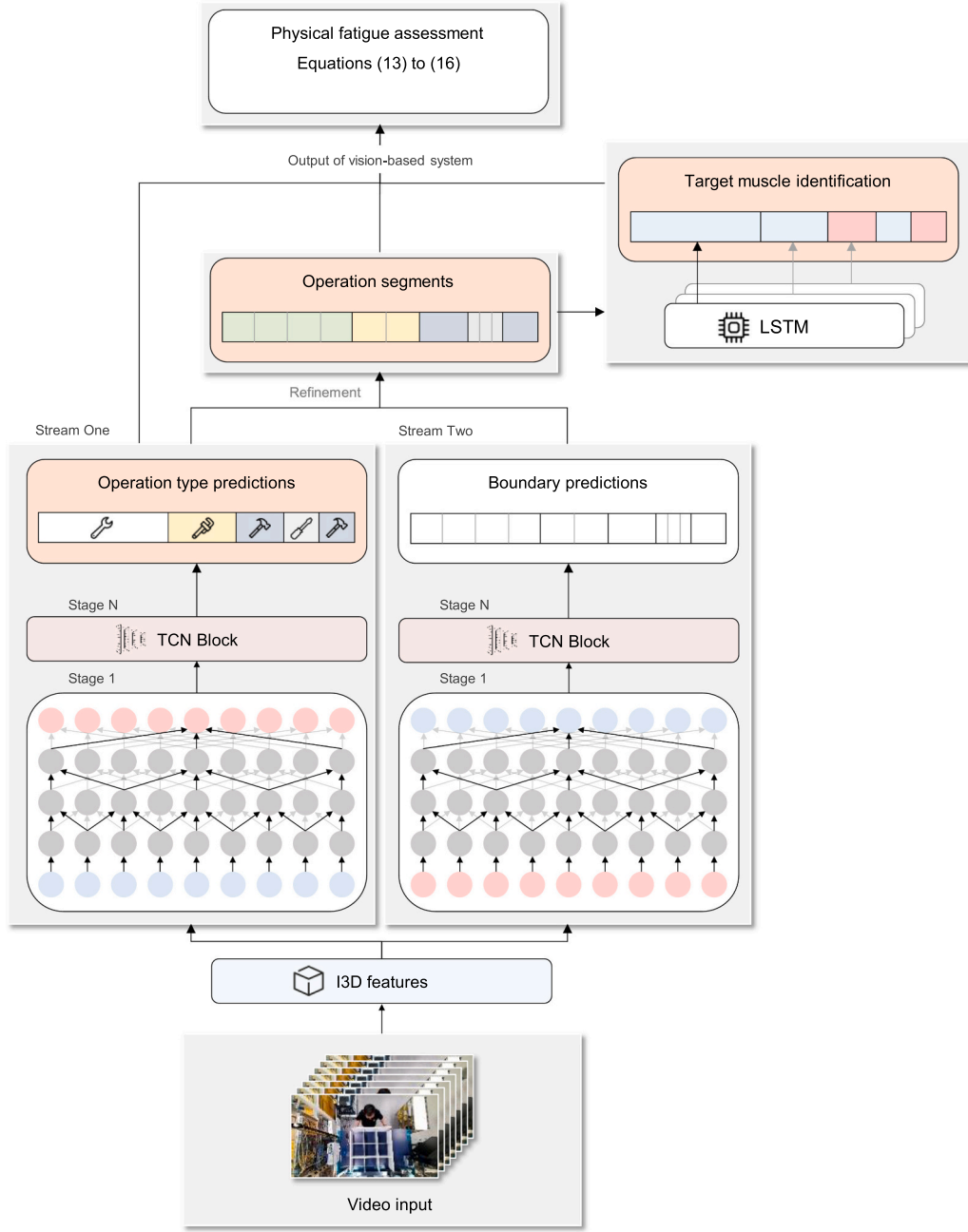


Fig. 5. Overview of our MS-TCN algorithm with an LSTM for operation segmentation and target arm recognition.

operation segment. For example, a task consisting of four screwing operations would have each screw be an operation segment. To address the problem of separating continuous repeated operations, a boundary-aware dual-stream MS-TCN algorithm is built upon from our previous research in [33]. Our algorithm uses two streams to predict the operation type and the operation repetition boundaries, illustrated in Fig. 5. Firstly, Inflated 3D ConvNet (I3D) features [34], which capture spatio-temporal information by applying 3D convolutional filters to the video frames, are extracted for each frame of the assembly process input video as $X = x_1, \dots, x_T \in \mathbb{R}^{T \times D}$, where T denotes the number of frames of the input video, and D denotes the dimension of the feature. Next, each stream takes X as the input and output framewise operation type predictions, $O = o_1, \dots, o_T \in \{0, 1\}^{T \times D}$, and boundary predictions, $B = b_1, \dots, b_T \in \{0, 1\}^T$. Then, boundary predictions are used to refine the operation type predictions to get the final operation segments. Each of the streams is trained separately.

In the operation type prediction stream, we use MS-TCN. Each TCN block operates on the video's entire temporal resolution, allowing the TCN block to capture long-term dependencies. The multi-stage architecture refines the prediction results stage-by-stage. The output layer of the prediction stream is a 1D convolutional layer followed by a SoftMax layer to predict framewise operation type probabilities $Y = y_1, \dots, y_T \in (0, 1)^{T \times C}$, where C is the number of operation classes. The loss function, referring to [27], combines a truncated mean squared error (T-MSE) and a classification loss. We define this as:

$$L_{ops} = L_{cls} + \lambda L_{T-MSE}; \quad (1)$$

$$L_{cls} = \frac{1}{T} \sum_t -\log(y_{t,c}); \quad (2)$$

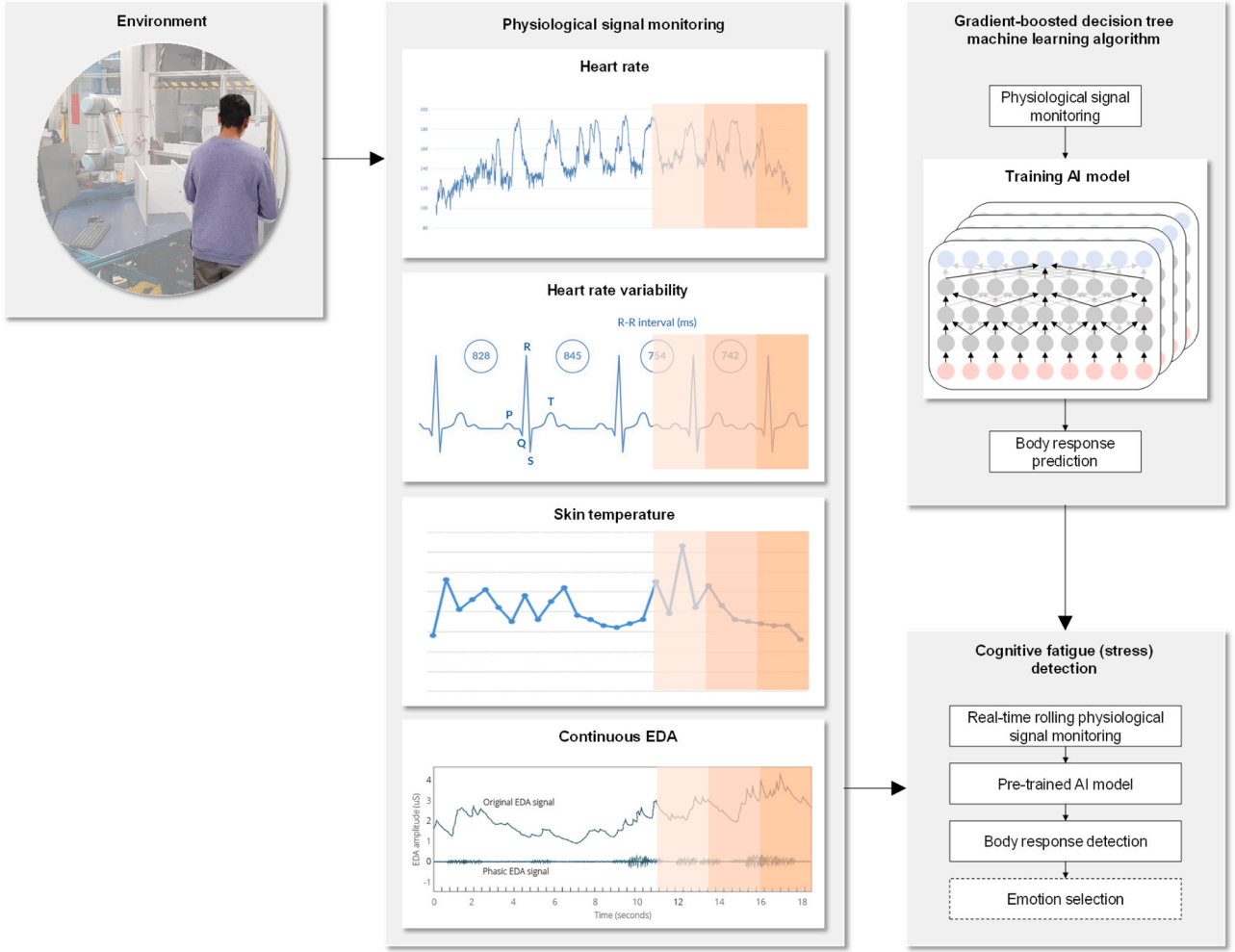


Fig. 6. Conceptual overview of the commercial cognitive fatigue monitoring and assessment tool using the Fitbit Sense 2 wearable sensor.

$$L_{T-MSE} = \frac{1}{TC} \sum_{t,c} \tilde{\Delta}_{t,c}^2; \quad (3)$$

$$\tilde{\Delta}_{t,c} = \begin{cases} \Delta_{t,c}, & \Delta_{t,c} \leq \text{thresh}, \\ \text{thresh}, & \text{otherwise}; \end{cases} \quad (4)$$

$$\Delta_{t,c} = |\log(y_{t,c}) - \log(y_{t-1,c})|; \quad (5)$$

Where $y_{t,c}$ is the probability of class c at time t . An argmax function is applied to map Y to O .

The second stream, which predicts operation boundaries, follows the same network architecture as the first stream, but the output is the framewise boundary probabilities $P = p_1, \dots, p_T \in (0, 1)^T$. The loss function L_{bps} is obtained by replacing the $y_{t,c}$ in L_{ops} with p_t , the boundary probability at time t . An argmax function is applied to map P to B .

Lastly, we sampled α I3D features from the original input for each operation segment and fed them into an LSTM network to classify the target muscle (either left or right arm in upper-body assembly) performing each operation segment.

The fundamental concept behind Long Short-Term Memory (LSTM) networks is their use of specialized gates to manage the flow of information, allowing them to retain important information over extended periods. In our study, we implement a well-known LSTM framework as outlined in [35]. At each time step τ , the LSTM updates its hidden state δ_τ using the current input v_τ , the previous hidden state $\delta_{\tau-1}$, and three gates: the input gate κ_τ , the forget gate q_τ , and the output gate ρ_τ . It also

utilizes a memory cell ς_τ . The equations governing these updates are as follows:

$$\kappa_\tau = \sigma(M_\kappa v_\tau + N_\kappa \delta_{\tau-1} + \omega_\kappa) \quad (6)$$

$$q_\tau = \sigma(M_q v_\tau + N_q \delta_{\tau-1} + \omega_q) \quad (7)$$

$$\rho_\tau = \sigma(M_\rho v_\tau + N_\rho \delta_{\tau-1} + \omega_\rho) \quad (8)$$

$$\varsigma_\tau = q_\tau \odot \varsigma_{\tau-1} + \kappa_\tau \odot \tanh(M_\varsigma v_\tau + N_\varsigma \delta_{\tau-1} + \omega_\varsigma) \quad (9)$$

$$\delta_\tau = \rho_\tau \odot \tanh(\varsigma_\tau) \quad (10)$$

In these equations, M , N , and ω are parameters learned during the training process, σ represents the sigmoid activation function, and $\odot$ indicates element wise multiplication.

The LSTM processes sequential data in chronological order. The output from the final time step is used to make the prediction via a linear regression layer, as shown in:

$$\bar{Y}_\tau = M_r \delta_{T_r} \quad (11)$$

In this formula, M_r is a weight matrix and T_r represents the terminal step. To train the model, we compare the predicted target arm value $\bar{Y}_\tau$ with the actual value Y using the MSE loss function:

$$\text{loss} = \frac{1}{N} \sum_{\tau=1}^N (\bar{Y}_\tau - Y_\tau)^2 \quad (12)$$

This loss function aids the LSTM in learning to make accurate predictions from the sequence of inputs.

4.1.2. Physical fatigue profile assessment

From our previous work [23], we assume that any assembly operation involves a combination of fatigue-inducing motion and non-fatigue-inducing motion, such as manipulating an object with a significant load versus reaching for said object. Miniscule physical recovery periods that occur during non-fatigue-inducing motions exhibit negligible effects on the localized muscle fatigue of an operator. Consequently, the presence of both fatigue-inducing and non-fatigue-inducing motions implies that merely counting the time taken for each task versus the number of repetitions may not provide an accurate measure of fatigue accumulation. Specifically, tasks with inflated times could, in fact, result in less fatigue accumulation if the extended duration includes a higher proportion of non-fatigue-inducing motions. This nuanced understanding highlights the need for a more detailed analysis of the nature of the motions involved in each task to accurately assess physical fatigue. Therefore, the fatigue of an operator p can be assessed by fatigue-inducing motions of an assembly task that involve an operation type, task load W_e , operation repetitions z , and is personalized to the operator's muscle q strength $S_{p,q}$. We can derive a task's relative load $W_{q,e}$ (for any task e and muscle q) by correlating the task's load with an operator's muscle strength $S_{p,q}$ as shown in Eq. (13). Our physical fatigue profile confirms linear relationships between the relative task load, repetitive operations, and fatigue accumulation. The physical fatigue profile equations are presented:

$$W_{q,e} = \frac{W_e}{S_{p,q}} \times 100\%. \quad (13)$$

$$m' = d \times z, \quad m' \in [0, 100\%] \quad (14)$$

$$d = \varepsilon_1 W_{q,e} + \varepsilon_0, \quad W_{q,e} \in [W_{low}, W_{up}]. \quad (15)$$

$$m' = (\varepsilon_1 W_{q,e} + \varepsilon_0)z, \quad m' \in [0, 100\%], \quad W_{q,e} \in [W_{low}, W_{up}]. \quad (16)$$

The fatigue profile as a percentage m' is linear to the number of repetitive operations z by the rate of fatigue d . This rate of fatigue d is linear to relative task load $W_{q,e}$ between the upper W_{up} and W_{low} , where W_{up} and W_{low} are defined as the upper and lower relative task loads specific to that particular operation type. The rate of fatigue linear coefficients is ε_1 and ε_0 . Fatigue accumulation across the number of repetitive operations is linear, as shown by Equation (14). Therefore, after obtaining a few values for d at different $W_{q,e}$ (preferably at the upper and lower relative task load bounds) using Eq. (15), we can apply two-point linear extrapolation to determine the linear coefficients ε_1 and ε_0 . To ensure statistical sufficiency in estimating these parameters, each operation type should be taken to failure for a particular relative task load $W_{q,e}$ at least five times and averaged out. Finally, with all the relevant variables predetermined, and the assembly process information established using our video understanding algorithm, we can determine the physical fatigue response to any task as a percentage automatically from the assembly video using Eq. (16).

4.2. Cognitive fatigue monitoring and assessment

This study employs the Fitbit Sense 2 wearable sensor for real-time cognitive fatigue detection through body responses, utilizing its commercial data collection and stress tracking algorithm. The lightweight and non-restrictive nature of the sensor, along with its diverse features beyond cognitive fatigue assessment, enhances its long-term appeal to operators. For these reasons, we incorporate the wearable sensor into our HHRC system.

Fig. 6 presents a conceptual overview of the Fitbit Sense 2 wearable sensor solution for cognitive fatigue tracking, comprising four modules:

Table 1

Task and resource information for the bookshelf assembly including execution times for one operator and robot.

Task index	Operation type	Task load	Operation repetitions	Operator A exec times	Robot exec times	Agent mode
1	MC	2.8	1	22	30	H/R
2	MC	1.8	2	34	45	H/R
3	SS	N/A	2	127	150	H/R
4	MC	2.8	1	22	30	H/R
5	MC	1.8	2	34	45	H/R
6	SS	N/A	2	127	150	H/R
7	MC	2.8	1	22	30	H/R
8	MC	1.8	2	34	45	H/R
9	SS	N/A	2	127	150	H/R
10	HD	4.0	4	31	40	H/R
11	SC	N/A	2	42	N/A	H
12	IC	6.4	1	39	60	H/R
13	HD	4.0	4	31	40	H/R
14	SC	N/A	2	42	N/A	H
15	IC	6.4	1	39	60	H/R
16	RC	9.6	1	21	21	H&R
17	MC	2.8	1	22	40	H/R
18	HD	4.0	4	31	45	H/R
19	SC	N/A	2	45	N/A	H
20	IC	6.4	1	39	45	H/R
21	RC	11.0	1	25	25	H&R
22	MC	2.8	1	22	40	H/R
23	SS	N/A	8	254	400	H/R
24	RC	12.4	1	31	31	H&R
25	MC	2.8	1	22	40	H/R
26	SS	N/A	8	254	400	H/R

the physical environment, real-time physiological signal monitoring, a pre-trained AI model for quantifiable stress detection, and the cognitive fatigue detection output. The sensor, worn on the operator's wrist, continuously monitors physiological signals in real-time, including heart rate, heart rate variability, skin temperature, and electrodermal activity (EDA) signals. These signals are collected by the Fitbit sensor's integrated biosensors, which provide high-resolution, continuous data streams [36]. The collected signals are then processed within a rolling window and fed into the AI model. Fitbit uses a gradient-boosted decision tree algorithm to handle both classification and regression tasks. This pre-trained AI model, developed using extensive clinical data, is designed to recognize patterns associated with stress and fatigue. By analyzing physiological signals collected from the device, the model consolidates this data to identify body responses indicative of stress [18]. Body responses, while quantitative measures of stress, can result

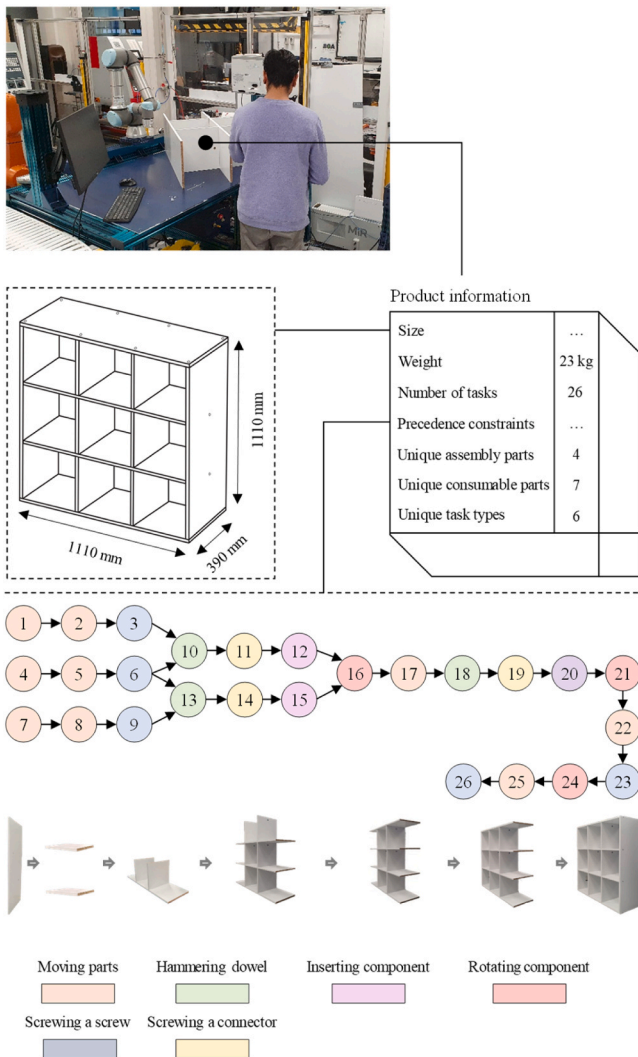


Fig. 7. Bookshelf assembly performed by one operator and one robot team including precedence information.

from various emotional responses, both negative and positive. For instance, intense emotions, whether negative or positive, can trigger body responses; however, positive emotions may not be classified as cognitive fatigue. An optional step occurs when the sensor detects a body response assumed to be cognitive fatigue: the emotion selection step, allowing the operator to log their current mood. Future advancements may incorporate pattern recognition algorithms to identify trends in an operator's working patterns, enabling the autonomous scheduling of tasks or jobs between agents to mitigate negative emotions and optimize workflow.

At the system level, the HHRC system is expected to dynamically allocate immediate rest periods for operators who become cognitively fatigued. The system should reorganize the remaining tasks among the available agents to ensure productivity is unaffected, enhancing operator health and comfort.

5. Case study

A case study was undertaken to validate the proposed methods in this study. The case study involved a 3×3 cube storage shelf, assembled under human-robot collaboration by a single operator and a single robot. Detailed task and resource information, including the task execution times for each agent required for task scheduling, are illustrated in Table 1.

In Section 5.2, to assess physical fatigue, we employed our boundary-aware MS-TCN with LSTM to analyze the impacts caused by each task in the bookshelf assembly. This approach allowed us to visually collect all key parameters necessary for physical fatigue assessment. A real-world product assembly provides a more illustrative demonstration of our proposed video understanding techniques in extracting relevant assembly process information for physical fatigue assessment.

Then in Section 5.4, we further explored different task allocation and scheduling scenarios for the bookshelf assembly under human-robot collaboration. This exploration aimed to optimize both the operator's physical fatigue levels and the assembly cycle time.

Additionally, In Section 5.3, we present body response detection data from a pilot study featuring one participant for cognitive fatigue assessment and monitoring, demonstrating the potential inclusion of this tool in our system to enhance overall fatigue management.

5.1. Human-robot collaborative assembly

The bookshelf product information and task precedence are highlighted in Fig. 7.

Table 1 shows detailed task and resource information, including task execution times for each agent.

5.2. Physical fatigue monitoring experiments and results

We obtained data showcasing the assembly process from a single operator constructing a bookshelf at a workstation. The operator assembled the bookshelf 20 times, resulting in 20 full-length videos, with each assembly consisting of 26 tasks. Of these, 16 videos were randomly selected as the training set, while the remaining four videos were designated as the testing set. In employing this dataset, we acknowledge the potential risk of overfitting to the desired results, given the limited number of training examples. Overfitting occurs when a model learns the specific details and noise in the training data to the extent that it performs poorly on new, unseen data. To mitigate this risk, we employed several strategies. Firstly, random selection of training and testing sets ensures that our model is exposed to a diverse range of data during training. Additionally, we implemented techniques such as cross-validation and regularization to enhance the model's generalizability. It is important to note that this study serves as a proof-of-concept. While the results are promising, a larger dataset would be ideal for real-world applications. Training on a more extensive dataset would provide a more comprehensive understanding of various task scenarios and operator behaviors, thereby improving the robustness and accuracy of the model in practical settings. Consequently, future work should focus on collecting and utilizing a more extensive dataset to validate and refine the proposed methods further. Each video encompasses nine distinct operation types and 64 operation segments. The labeling process for these videos involved recording the time stamps for the start and end of each operation segment and each task. Additionally, each segment and task were annotated with the corresponding operation type and the target muscle group, indicating whether the dominant or non-dominant arm was used. This meticulous labeling ensures precise identification and analysis of the various operations and muscle groups engaged during the assembly process, providing a robust foundation for our MS-TCN and LSTM networks.

To validate our proposed approach, we present our boundary-aware MS-TCN with LSTM for operation segmentation and target arm classification. For our network, each stream is stacked in four stages, with each stack containing ten dilated residual convolution layers. The dilation factor is doubled at each layer, and the dropout rate is 0.5 for each layer. Each layer has 64 filters, and the filter size is 3. For the loss function, $\lambda = 0.15$ and $thresh = 4$. We sample $\alpha = 16$ I3D features evenly for each operation segment and feed them into a two-layer LSTM network. We then compared our algorithm's performance in operation segmentation against other algorithms, including MS-TCN, DTGRM,

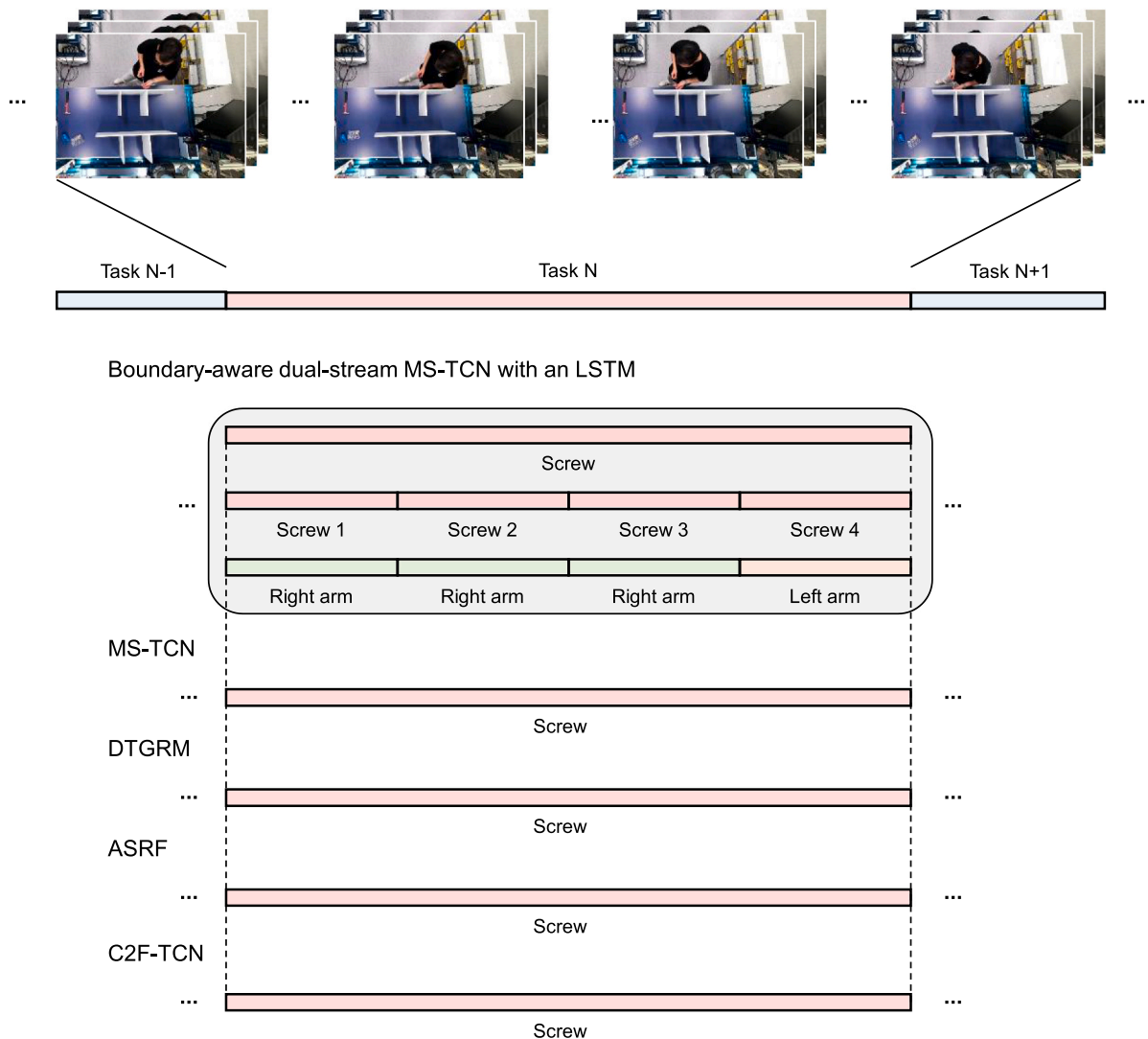


Fig. 8. Illustration of operation segmentation results comparing our boundary-aware dual-stream MS-TCN with an LSTM algorithm against the MS-TCN, DTGRM, ASRF, and C2F-TCN.

Table 2
Performance of our algorithm at operation segmentation and identifying the target arm performing each task.

	Frame-wise accuracy (operation type recognition)	Operation segments per video	Target arm accuracy
Ours	99.97 %	64.0/64.0	100 %

Table 3
Performance of our algorithm compared to similar action segmentation algorithms, including MS-TCN, DTGRM, ASRF, and C2F-TCN.

	Frame-wise accuracy (operation type recognition)	Operation segments per video
MS-TCN	99.87 %	26.0/64.0
DTGRM	99.81 %	26.5/64.0
ASRF	99.69 %	25.5/64.0
C2F-TCN	99.80 %	26.0/64.0
Ours	99.97 %	64.0/64.0

ASRF, and C2F-TCN.

Table 2 showcases our algorithm’s base results against three key objectives: frame-wise accuracy (operation type recognition), operation segmentation, and target arm accuracy. These three results show near-optimal performance in extracting the relevant assembly information for physical fatigue assessment.

Our algorithm performance, compared to state-of-the-art action segmentation algorithms, is evaluated based on frame-wise accuracy and the effectiveness of distinguishing consecutive repeated operations as shown in Table 3. Unlike other algorithms, the LSTM component in our algorithm uniquely identifies the target arm for each segment of the operation. The four different algorithms are unable to identify all operation segments in the video correctly. Specifically, repeated

Table 4
Fatigue parameters for each operation type.

Operation type	β_1^o	β_o^o
Moving parts (MC)	0.086	1.143
Hammering dowel (HD)	0.167	−0.001
Inserting component (IC)	0.296	−3.335
Rotating component (RC)	0.190	0.477
Screwing a screw (SS)	N/A	2.000
Screwing a connector (SC)	N/A	5.000

Table 5

Detailed assembly process information with task breakdown including each task's physical fatigue response. The assembly process information was sourced from the input video using our boundary-aware MS-TCN with an LSTM algorithm. The target arm and muscle strength associated with each task in this table was the operator's dominant arm for simplicity.

Task index	Operation type	Task load	Operation segments	Operator execution time	Fatigue parameter	Fatigue response
1	MLP	2.8	1	22	8.33	8.33
2	MSP	1.8	4	34	4.00	16.00
3	SS	N/A	2	127	2.00	4.00
4	MLP	2.8	1	22	8.33	8.33
5	MSP	1.8	4	34	4.00	16.00
6	SS	N/A	2	127	2.00	4.00
7	MLP	2.8	1	22	8.33	8.33
8	MSP	1.8	4	34	4.00	16.00
9	SS	N/A	2	127	2.00	4.00
10	HD	4.0	4	31	5.00	20.00
11	SC	N/A	2	45	5.00	10.00
12	IC	6.4	1	39	10.00	10.00
13	HD	4.0	4	31	5.00	20.00
14	SC	N/A	2	45	5.00	10.00
15	IC	6.4	1	39	10.00	10.00
16	R1	9.6	1	21	8.33	8.33
17	MLP	2.8	1	22	8.33	8.33
18	HD	4.0	4	31	5.00	20.00
19	SC	N/A	2	45	5.00	10.00
20	IC	6.4	1	39	10.00	10.00
21	R2	11.0	1	25	10.00	10.00
22	MLP	2.8	1	22	5.00	5.00
23	SS	N/A	8	254	2.00	16.00
24	R3	12.4	1	31	11.10	11.10
25	MLP	2.8	1	22	8.33	8.33
26	SS	N/A	8	254	2.00	16.00

Table 6

Outline of body responses data over two weeks.

Day	1	2	3	4	5	6	7
Number of responses	3	2	3	4	3	1	2
Average response duration (mins)	5	3.5	6	4.5	6	2.5	4
Day	8	9	10	11	12	13	14
Number of responses	3	4	4	4	3	2	1
Average response duration (mins)	6	6.5	4.5	5	5	3.5	3

operations for a single task. On the other hand, algorithms such as MS-TCN cannot segment repeated operations for the same task. Our algorithm superiority is validated by considering the ground truth as 64 segments per video. Fig. 7 shows a visual representation of our algorithm's performance against the MS-TCN.

After the above video understanding techniques, we obtain the operation type, segments, and the target arm performing each task. Using the physical fatigue assessment equations defined in Section 4.1.2, each task's fatigue response can be obtained, as shown in Table 5. The fatigue parameters for each operation type are shown in Table 4 and obtained through the methods discussed in Section 4.1.2 via a set of pilot experiments not explicitly stated in this paper. Note, this table represents the fatigue response for the operator's dominant arm as the target muscle, with a muscle strength $S_{p,q} = 18\text{kg}$. Note, the operator's non-dominant arm muscle strength is evaluated as $S_{p,q} = 16\text{kg}$ used for task scheduling and allocation. More information into the methods used to determine the fatigue parameters can be found in [23].

5.3. Cognitive fatigue monitoring results

We present data showcasing the detection of body responses indicative of stress (cognitive fatigue) over a 24-hour period, utilizing the Fitbit Sense 2 wristband sensor. The body responses occur naturally as the operator becomes cognitively fatigued throughout the day. The spikes in physiological signals, which include heart rate, heart rate variability, skin temperature, and electrodermal activity (EDA), are not instantaneous but persist for extended durations, typically lasting

between 2 and 9 min. Over the two-week data collection period, body response detection was triggered between 1 and 4 times daily, with an average duration of 2.5 to 6.5 min per instance, as detailed in Table 6. These observed spikes in physiological signals align closely with the ground truth of the operator's reported experiences of cognitive fatigue, validating the accuracy of the sensor data. However, there were seven exceptions where the spikes were caused by intense physical activity rather than cognitive fatigue. Additionally, among all the spikes in physiological signals driven by cognitive factors, all but two were identified as cognitive fatigue linked to negative emotions based on the operator's emotion selection at the time of the response. The remaining two instances were associated with positive emotions and were not considered cognitively fatiguing.

The detected body responses serve as indicators of potential cognitive fatigue, as spikes in these physiological signals are widely documented in the literature as markers of cognitive fatigue [11,22]. The monitoring of these signals allows for the assessment of cognitive fatigue in a binary fashion, categorizing the operator as either cognitively fatigued or not. With the current technologies, it is not yet possible to assess or monitor the degree of cognitive fatigue while remaining unintrusive and comfortable to the operator. Furthermore, there is no automated response to an operator being identified as cognitively fatigued. Instead, it is recommended that operators who exhibit signs of cognitive fatigue take a break. Consequently, the task scheduling system must possess sufficient robustness to reorganize tasks dynamically to accommodate these breaks. It should be noted, however, that the development of a dynamic scheduling algorithm to address this need is beyond the scope of this research.

It is important to acknowledge the limitations inherent in physiological signal tracking, as these signals can be confounded by intense physical activity. Exercise can significantly elevate heart rate and heart rate variability, increase skin temperature, and affect EDA responses, potentially leading to false positives in cognitive fatigue detection. However, within the manufacturing environment, strenuous physical activity is minimized. Specifically, our system integrates a physical fatigue assessment module that identifies and mitigates significant physical exertion before it can interfere with the physiological signals used to

Table 7
GA algorithm parameters.

Parameter	Value
Epochs	5000
Population	500
CXPB	0.5
MUTPB	0.2
MTR	0.1
Objectives	2
Partitions	2

detect cognitive fatigue. This approach ensures that the physiological data primarily reflect cognitive stress rather than physical activity, enhancing the accuracy of cognitive fatigue detection in an industrial setting.

5.4. Fatigue-sensitive task scheduling

We present a sample task scheduling application involving the bookshelf assembly introduced in Section 5.1 under human-robot collaboration. The bookshelf task precedence information is described in Fig. 7, where each task’s operation type is highlighted. The resulting task breakdown including agents’ capabilities are shown in Table 1.

Based on the precedence information and task breakdown, the human-robot task scheduling model can be formulated. Here, we set the optimization objectives to minimize the maximum physical fatigue and the cycle time, as conflicting multi-objectives that are solved for using a Genetic Algorithm (GA) metaheuristic. The metaheuristic algorithm parameters are detailed in Table 7.

After the optimization, an optimal solution is obtained as shown in Table 8. The total cycle time observed in this solution is 1614.6 s and the maximum fatigue of the operator is 45.5 % which includes both their dominant and non-dominant arms. Note, tasks assigned to an operator’s non-dominant arm have an increased execution time of x1.2 to reflect laterality differences between their arms.

Fig. 9 showcases the fatigue profile of the operator’s dominant arm over the first 13 tasks in the bookshelf assembly. In particular, we highlight the physical fatigue accumulation and recovery process, where recovery is modeled using the Learning-Forgetting-Fatigue-Recovery (LFFR) model dependent on the recovery period and the rate of recovery parameter which is set to 0.003 as per findings in the original study [37]. Furthermore, we designate a cognitive fatigue threshold, which is triggered by the body response detection following the discussions in Section 5.3. As this occurs, the system provides a rest period to the worker, reassigning future tasks where required.

6. Conclusions

This study introduces significant advancements in HHRC and fatigue monitoring in manufacturing settings. Central to our contribution is a novel HDT framework that integrates both physical and cognitive fatigue modeling of operators during the assembly process. This comprehensive approach enhances the understanding of operator states and performance, crucial for developing truly human-centric HRC systems.

Our primary innovation is a vision-based boundary-aware operation segmentation algorithm. This non-invasive method overcomes the drawbacks of traditional fatigue monitoring, which relies on intrusive sensors, by classifying repetitive operations performed by either arm using video analysis. Coupled with an existing localized muscle fatigue profile, we present a personalized, non-contact physical fatigue

assessment method. This integration creates a pathway for real-time fatigue monitoring without interrupting the operator’s workflow, making it highly practical for industrial applications.

By incorporating wearable sensors for cognitive fatigue detection, our study provides a comprehensive view of operator state, addressing both physical and mental fatigue in manufacturing environments. This multifaceted approach to fatigue monitoring and assessment marks a significant step towards more intelligent, adaptive, and operator-focused HRC systems.

We validated our approach through extensive evaluation in a real-world assembly scenario involving 26 tasks and six distinct operation types. This thorough testing confirms the feasibility and effectiveness of our methods in complex, varied assembly processes typical of modern manufacturing environments. Additionally, we demonstrate the potential for fatigue-sensitive task scheduling in HHRC by optimizing both cycle time and operator fatigue levels, highlighting how our HDT framework can improve productivity and operator well-being.

In conclusion, our contributions advance human-centric manufacturing, providing a robust foundation for future research in multi-modal fatigue monitoring, dynamic task allocation, and proactive fatigue management in industrial settings. Our work addresses current challenges in HHRC and paves the way for more sophisticated, human-centered manufacturing approaches in the era of Industry 5.0.

Our methods demonstrate considerable validity within this application space. However, additional work is necessary to address the challenges associated with real-time physical fatigue monitoring and assessment. Future research may explore the integration of our vision-based system with other forms of data, creating a multi-modal system. Such a system could incorporate sensor data, physiological signals, and environmental factors while maintaining a non-intrusive and comfortable solution for the operator over long periods.

Moreover, our current system evaluates physical and cognitive fatigue by analyzing different data streams in parallel, providing a holistic view of the operator’s condition. The physical fatigue assessment is comprehensive, but the cognitive fatigue assessment is currently limited to determining whether the operator is cognitively fatigued or not. In the future, it may be possible to integrate cognitive fatigue assessments more deeply into the system, allowing for dynamic task adjustments and more proactive fatigue management.

Future research should also consider integrating the system with

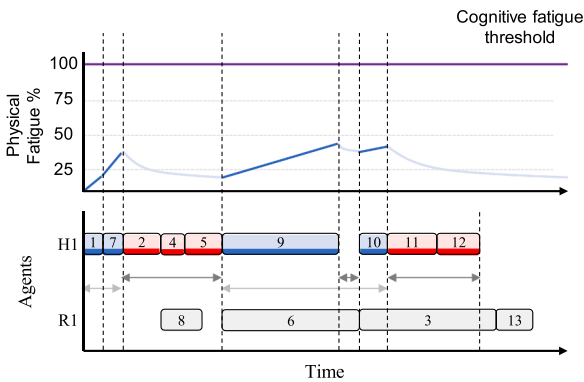


Fig. 9. Fatigue-sensitive scheduling solution with the first 13 tasks of the bookshelf assembly. Tasks highlighted in blue are assigned to the operator’s dominant arm and tasks highlighted in red are assigned to the operator’s non-dominant arm.

Table 8
Optimal fatigue-sensitive scheduling solution minimizing cycle time and fatigue.

Fatigue objective	Dominant arm tasks	Non-dominant arm tasks	Robot tasks	Cycle time (s)	Maximum Fatigue (%)
Optimal solution	1,7,9,14,10,19,21,25,26	2,4,5,11,12,16,22,24	8,6,3,13,15,16,17,18,19,20,21,23,24	1614.6	45.5 %

Human-Robot Collaboration (HRC) task scheduling for human-centric task optimization across multiple operators and robots. This enhancement would enable the HRC system to monitor and mitigate physical fatigue accumulation for each operator individually, thereby improving overall productivity and worker well-being.

CRedit authorship contribution statement

Yuqian Lu: Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Investigation, Funding acquisition, Conceptualization, Methodology. **Saahil Chand:** Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Hao Zheng:** Writing – original draft, Validation, Methodology.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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